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HOSPITAL OPERATION

Performance Report

AN INSIGHT INTO THE DATA

PRESSURE FLOW OF PATIENTS

CONGESTION AREAS

OPERATIONAL INSIGHTS FOR LEADERSHIP

EXECUTIVE SUMMARY

The analysis reveals that patient flow pressure within the hospital is driven less by clinical complexity and more by operational inefficiencies, particularly in discharge processes and care transitions. While daily admissions remain relatively stable, the system experiences periodic surges, especially during early morning and evening hours. These peaks, occurring around shift transition periods, create intermittent congestion that places pressure on staffing, triage, and bed availability.

Despite this steady inflow, the average length of stay (LOS) remains consistently around four days across departments, diagnoses, and discharge outcomes. This uniformity is notable because it suggests that LOS is not strongly influenced by patient condition, severity, or demographic factors. Instead, it points to a system governed by standardized processes or discharge timelines, which may not fully align with individual patient needs. As a result, inefficiencies in patient flow are likely systemic rather than clinical.

The most significant bottleneck emerges at the point of discharge, particularly for patients requiring post-acute care such as rehabilitation or skilled nursing. A substantial proportion of patients depend on these external facilities, making discharge timing heavily reliant on external capacity. This creates delays in freeing up beds, leading to downstream congestion that compounds during periods of high admission volume. Older patients, in particular, contribute to this challenge, not by staying longer, but by requiring more complex discharge coordination.

Further analysis provides additional nuance to LOS patterns. While older patients do not exhibit the longest stays, younger adult patients show slightly higher LOS despite lower admission volumes. This appears to be driven by specific diagnoses such as normal delivery, urinary tract infections (UTIs), and appendicitis, all of which are associated with moderately longer recovery periods. These conditions often involve structured care pathways, such as post-surgical monitoring or maternal recovery, that extend hospital stays in a predictable manner.

A similar pattern is observed in urgent admissions, which, although fewer in number, have the highest average LOS. This can be attributed to diagnoses like UTIs, appendicitis, and hypertension, which require careful stabilization and monitoring but are not immediately critical. These cases tend to fall between emergency and elective pathways, often resulting in longer inpatient durations due to diagnostic uncertainty, treatment progression, or delays in achieving discharge readiness.

At the departmental level, Orthopedics and Pediatrics experience the highest admission volumes, indicating localized pressure points within the hospital. When combined with even slightly elevated LOS, these departments are more likely to face capacity strain over time. However, since LOS variation across departments is minimal, the constraint is driven more by volume intensity than inefficiency.

Overall, the key operational insight is that the hospital's primary constraint lies not in treating patients, but in moving them efficiently through and out of the system. The combination of steady admissions, bi-modal demand peaks, reliance on external discharge pathways, and standardized LOS patterns creates a system where small inefficiencies quickly accumulate into significant congestion.

For leadership, the priority should be to optimize discharge planning and strengthen coordination with post-acute care facilities, while also aligning staffing and operational workflows with actual admission patterns. Addressing these areas will have the greatest impact on improving patient flow, reducing bottlenecks, and enhancing overall system responsiveness.